

SOFTWARE REQUIREMENTS SPECIFICATION

Inventory Management System

1. Introduction & Overview

1.1 Project Purpose

The purpose of this project is to design and develop a digital management system for small-to-medium retail businesses in the building materials sector. The system aims to replace manual, paper-based operations by automating billing, GST calculation, inventory tracking, and customer record management.

The project focuses on improving operational efficiency, reducing calculation errors, minimizing data redundancy, and providing structured access to product and sales information.

1.2 System Overview

The system is a web-based retail management platform that automates:

- Billing & Invoice Generation
- GST Calculation
- Inventory Management
- Customer Record Management
- Product Catalog Management

The system improves accuracy, efficiency, and accessibility compared to traditional manual workflows.

2. User Roles & Characteristics

2.1 System Actors

The system supports multiple user roles with different access levels to ensure security and proper management of operations.

1. Admin (Owner)

The Admin has complete control over the system and is responsible for managing overall business operations within the platform.

Responsibilities:

- Manage user accounts and assign roles.
- Add, update, and delete products in the inventory.
- Monitor stock levels and manage product categories.
- Configure GST rates and tax rules.
- Access sales reports, GST summaries, and business analytics.
- Monitor system activities and ensure proper data management.

2. Staff (Billing Operator)

Staff members are responsible for handling daily billing operations and assisting customers during transactions.

Responsibilities:

- Generate new sales invoices for customers.
- Search and select products from the product catalog.
- Enter or update customer details during billing.
- View previous invoices and transaction history.
- Update stock quantities during manual adjustments when permitted.

3. Viewer (Accountant)

The Viewer role is intended for accountants or auditors who require access to financial records without modifying system data.

Responsibilities:

- View invoices and transaction records.
- Access sales and GST reports for accounting purposes.
- Export reports for tax filing or financial analysis.
- Review business performance data without editing permissions.

3. Core Features Overview

3.1 Billing Management (BM)

The Billing Management module handles the complete invoice generation process and ensures GST-compliant billing.

Key Functions:

- Generate new sales invoices for customers.
- Automatically calculate GST based on configured tax rates.
- Display tax breakdown including CGST, SGST, or IGST.
- Print or download invoices in digital format.
- Maintain invoice history for future reference.
- Cancel or modify invoices when required.

3.2 Inventory Management (IM)

The Inventory Management module ensures accurate tracking and control of product stock levels.

Key Functions:

- Add new products to the inventory.
- Edit or remove existing products.
- Categorize products based on material type (cement, steel, pipes, hardware, etc.).
- Track current stock levels in real time.
- Automatically update stock after each sales transaction.
- Identify low-stock items to prevent stock shortages.

3.3 Customer Management (CM)

The Customer Management module stores and manages customer information required for billing and reporting.

Key Functions:

- Register new customers in the system.
- Store customer contact details and GSTIN information.
- Maintain purchase history for each customer.
- Quickly search and retrieve customer records during billing.

3.4 Reporting & Analytics (RA)

The Reporting and Analytics module provides insights into business performance and financial transactions.

Key Functions:

- Generate monthly sales reports.
- Produce GST summary reports for tax filing.
- View inventory stock reports.

- Export reports in PDF or Excel format.

4. Functional Requirements

ID	Requirement Title	Description	Input	Process	Output	Priority
1	Generate New Sale Invoice (Billing Management)	Staff generates a new sales invoice for a customer.	Customer details, Product list, Quantity	Fetch product price; Calculate subtotal; Apply GST automatically; Generate total amount	Invoice generated with unique invoice number; Stock automatically updated	P0
2	GST Calculation (Billing Management)	System automatically calculates applicable GST (CGST, SGST, IGST).	Product price, GST %	$GST = (Product\ Price \times GST\ \%)$	Accurate tax breakdown displayed on invoice	P0
3	Add Product (Inventory Management)	Admin adds new product into the system.	Product name, category, price, GST %, stock quantity	Validate no duplicate product names	Product stored in system	P0
4	Auto Stock Update (Inventory Management)	Stock quantity automatically reduces after each sale.	Sale transaction data	Deduct sold quantity from available stock	Updated stock value	P0

5	Add Customer (Customer Management)	Add new customer details into the system.	Name, phone number, GSTIN (optional), address	Validate phone number format	Customer details saved	P0
6	Generate Sales Report (Reporting & Analytics)	System generates monthly sales reports.	Date range selection	Aggregate sales data	Report viewable and exportable (PDF/Excel)	P1

5. Non-Functional Requirements

5.1 Performance

- Page load time < 5 seconds
- API response time < 700 ms
- Support up to 500 concurrent users

5.2 Scalability

- Support up to 8,000 products
- Allow multi-user access

5.3 Availability

- 95.5% uptime SLA
- Maintenance during non-business hours

5.4 Security

- Role-based access control
- Encrypted passwords
- Secure login authentication
- Data protection for GST records